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The global state of cranioplasty practice following cranial decompression for traumatic brain injury: a provider survey☆

Sara Venturini[a,*], Saniya Mediratta[a], Tobias J. Adams[a], Harry Mee[a], Bhagavatula Indira Devi[b], Tariq Khan[c], Andres M. Rubiano[d,e], Franco Servadei[f], Adel Helmy[g], Angelos Kolias[a], Peter J. Hutchinson[a]

[a] Global Health Research Group on Acquired Brain and Spine Injury, University of Cambridge, Cambridge, United Kingdom
[b] Department of Neurosurgery, National Institute of Mental Health and Neuro Sciences, NIMHANS, Bengaluru, Karnataka, India
[c] Department of Neurosurgery, North Western General and Research Hospital, Peshawar, Pakistan
[d] Neuroscience Institute, Universidad El Bosque, Bogota, Colombia
[e] INUB Meditech Group, Meditech Foundation, Cali, Colombia
[f] Global Neurosurgery Programme, Besta Neurological Institute, Milan, Italy
[g] Division of Neurosurgery, Department of Clinical Neurosciences, University of Cambridge, Cambridge, United Kingdom

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ABSTRACT

Objective: For patients undergoing decompressive craniectomy for TBI, cranioplasty facilitates rehabilitation and reintegration into society. Cranioplasty practice is poorly documented globally. Challenges including lack of materials and technical skills disproportionately affect LMICs. This study evaluates global cranioplasty practice and the extent to which barriers preclude patients from accessing cranial reconstruction.

Methods: An international survey was disseminated to centres performing cranioplasty for TBI. Survey questions addressed baseline hospital information, cranioplasty indications, barriers and techniques, follow-up. Centres in HICs and LMICs were compared.

Results: 101 responses were received (86 individual institutions, 39 countries). Variation in practice was seen globally, and between HICs and LMICs. Autologous bone was the most common material used. Titanium and polymethylmethacrylate were the most used artificial implants. LMIC sites were more likely to store bone flaps in the patient's abdomen, while HICs had more access to 3D printing. Lack of infection and good neurological recovery were the commonest eligibility criteria. Cost of materials/operation and unavailable materials were the commonest barriers. Responders suggested easier access to cheaper materials would significantly improve access. Cranioplasty associated costs were higher than the country's GNI per capita in 7 cases. Less than 50 % of patients without cranioplasty had access to protective equipment. Less than a quarter of respondents stated patients had access to brain injury charities, and over 50 % believed stigma affects their patients.

Conclusions: Variation in cranioplasty practice was confirmed. Barriers limiting access were identified, specifically, availability of materials and operation-related costs. These findings can inform context-specific interventions to overcome current challenges.

1. Introduction

Traumatic brain injury (TBI) can cause intracranial hypertension through a variety of pathomechanisms.[1] To treat this, decompressive craniectomy (DC) is a common and effective operation involving removing a portion of the skull and opening the underlying dura mater.[2–4] For patients who survive, cranial reconstruction (cranioplasty) is usually indicated to repair the defect. In addition to restoring cerebral protection and improving cosmesis, cranioplasty can improve cerebral blood flow, neurological function,[5,6] and may aid physical and cognitive rehabilitation.[7] Possible complications include infection, hydrocephalus, intracranial haemorrhage and seizures.[8,9]

Cranioplasty is also performed following DC for indications other than TBI; primarily following ischaemic stroke but sometimes following

☆ Previous presentations: results of the study were presented at the Society of British Neurological Surgeons (SBNS) meetings.

* Corresponding author. Division of Neurosurgery, Department of Clinical Neurosciences University of Cambridge, Cambridge, CB2 0QQ, United Kingdom.
E-mail address: sv465@cam.ac.uk (S. Venturini).

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spontaneous intracranial haemorrhage. This study addresses cranioplasty performed following TBI.

There is variation in cranioplasty practice and the optimal material for reconstruction has yet to be agreed upon.[10,11] Synthetic materials can be used including the thermoplastics polymethylmethacrylate and polyetheretherketone; titanium in the form of plates, malleable meshes and 3D porous implants; and osteoconductive, porous hydroxyapatite. A common alternative is autologous bone, which can be stored in an abdominal pocket or undergo extracorporeal cryopreservation,[12] but has the potential issue of bone flap resorption.[13]

The timing from bone removal to cranioplasty is variable, largely owing to inconsistent definitions of "early cranioplasty" and mixed results from investigating associations between timing, complications[10,14,15] and neurological outcome.

To address variation in clinical practice within and between countries, the first consensus-based recommendations for cranioplasty were published in 2021.[10] The ability to offer cranioplasty is influenced by the available technical skills, neurosurgical facilities, and presence of institutional frameworks for treatment and follow-up. Challenges within these factors are likely greater in low- and middle-income countries (LMICs), yet the extent to which barriers preclude patients from accessing cranioplasty has not been formally investigated. Therefore, this study aims to evaluate global practice around cranioplasty for TBI, and identify barriers to access, as perceived by practicing professionals.

2. Methods

A cross-sectional study was performed through dissemination of an online survey (Supplementary Material 1). A summary of the domains addressed in the survey is presented in Table 1.

2.1. Survey design

The survey was designed based on recent cranioplasty literature, including the recent consensus statement,[10] to capture key aspects of care. It was created using Google Forms (Google) as this is free-to-access and accessible globally via smartphone or computer. The survey was piloted by the NIHR Global Health Research Group on Acquired Brain and Spine Injury to ensure questions were clear and relevant to clinicians working across a range of settings. Feedback was sought on survey content, structure, ease of completion.

2.2. Survey dissemination

A systematic *a priori* dissemination strategy was developed to capture responses across World Bank Income Groups and WHO Regions and was phased to enable response rate monitoring.

Phase 1 consisted of invitation directly via email. Respondents were selected from institutions across WHO Regions, aiming to obtain one response per neurosurgery centre. Invitation and participation logs were kept throughout. Members of the following groups were initially invited to participate.

Table 1
Summary of survey questions and sections.

Section	Question domains	Question numbers
Section 1: baseline details	Responder details	1–3
	Hospital information	4–10
Section 2: cranioplasty information	Initial cranial decompression surgery	11–15
	Cranioplasty indications, obstacles and techniques	16–35
	Follow-up and ongoing patient care	36–50

- NIHR Global Health Research Group on Acquired Brain and Spine Injury
- WFNS Global Neurosurgery steering committee
- Participating centres of the Global Neurotrauma Outcomes Study (GNOS) run by the NIHR GHRGABSI

The second phase expanded invitation through neurosurgical societies, including the WFNS newsletter. Phase three utilised dissemination via social media platforms.

2.3. Data analysis

Data was analysed using Microsoft Excel and R.[16,17] A descriptive representation of the data was used. Gross national income (GNI) was used to group countries into high, upper-middle, lower-middle, and low-income groups. Comparison between country income groups was performed using Fisher's Exact tests. Upper middle-, lower middle- and low-income countries were grouped into the LMIC group for statistical comparison due to low numbers in each category, when comparing groups according to the study variables. Statistics comparisons were then performed between the high-income country (HIC) and LMIC groups. A p-value of <0.05 indicated statistical significance.

The World Bank was used as the source for GNI and GDP (gross domestic product) data at country level, for 2021, the most recent year available.[18] Where multiple responses were provided for the same institution, discrepancies were identified and respondents individually contacted by email for clarification.

3. Results

101 responses were received. After removing duplicates ($n = 13$) and responses from students ($n = 2$), responses from 86 independent healthcare centres across 39 countries, spanning 5 continents, were used in subsequent analysis (except free-text questions seeking individual opinion, where all responses were included). The global distribution of participating centres is summarised in Fig. 1. Countries with more than two centres represented were the United Kingdom (12.8 % of institutions surveyed), Italy (8.1 %), Ethiopia and India (7.0 % each) and Brazil (5.8 %), Kenya (5.7 %), Germany (4.7 %), United States (4.7 %) and Malaysia (3.5 %). 71 institutions (82.6 %) were tertiary centres and 84 (95.5 %) were in urban settings.

Baseline information from participating institutions is available in Table 2. Participating centres were similar with regards to respondent profession, hospital type, setting and type of funding across income groups (Fisher's exact, all $p > 0.05$), but hospitals in lower middle- or low income-countries were statistically more likely to charge elective care or all care (elective and emergency) to patients compared to hospitals in high or upper middle-income countries (Fisher's exact, $p < 0.001$).

3.1. Cranioplasty indications

Most centres (55, 64 %) offered cranioplasty to all eligible patients following DC and the commonest eligibility criteria were lack of infection (66 centres, 77 % of total), good neurological recovery (53, 62 %), and DC bone flap size (31, 36 %). There was no significant difference between HIC and LMIC groups (Fisher's exact, $p > 0.05$).

Sixty-one centres (71 %) estimated performing between 1 and 5 cranioplasty procedures each month with more than half of all centres (44) estimating over 75 % of patients undergoing DC proceeded to cranioplasty, with HIC and LMIC groups not differing significantly ($p > 0.05$). All HIC sites reported performing between 1 and 10 cranioplasties per month, but LMIC centres exhibited more variation with 4 often performing none and 4 high-volume centres performing over 10 a month.

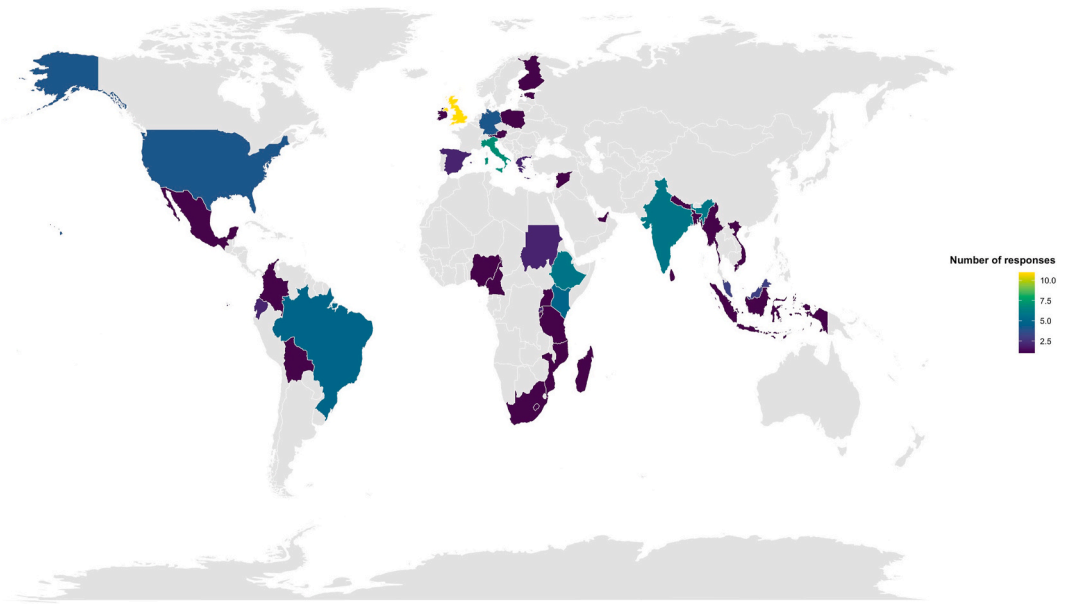

Fig. 1. Map displaying the geographical distribution of respondents. Different colours on the gradient represent greater number of responses per country. Responses were received from the following countries: Austria, Bangladesh, Bolivia, Brazil, Burundi, Cameroon, Colombia, Ecuador, Estonia, Ethiopia, Finland, Germany, Greece, India, Indonesia, Ireland, Italy, Kenya, Madagascar, Malaysia, Mexico, Mozambique, Myanmar, Nepal, Nigeria, Poland, Rwanda, Singapore, South Africa, Spain, Sri Lanka, Sudan, Syria, Tanzania, The Gambia, United Arab Emirates, United Kingdom, United States, Vietnam.

Table 2
Baseline information of the participating institutions, grouped by country income groups.

	Country income group				
	High income (n=37, 43 %)	**Upper middle income (n=14, 16 %)**	**Lower middle income (n=22, 26 %)**	**Low income (n=13, 15 %)**	Fisher's exact p value
Respondent profession					
Fully qualified neurosurgeon	25 (68 %)	11 (79 %)	18 (82 %)	8 (62 %)	0.697
Resident	12 (32 %)	3 (21 %)	3 (13 %)	4 (31 %)	
Anaesthetist/intensivist	0	0	1 (5 %)	0	
Other fully qualified surgeon	0	0	0	1 (7 %)	
Hospital setting					
Urban	35 (95 %)	14 (100 %)	22 (100 %)	13 (100 %)	0.777
Rural	2 (5 %)	0	0	0	
Hospital type					
Primary level	5	0	1	3	0.377
Secondary level	3	0	2	1	
Tertiary level	29	14	19	9	
Hospital funding					
Fully government funded	36 (97 %)	11 (79 %)	19 (86 %)	12 (92 %)	0.072
Private: insurance funded	1 (3 %)	0	1 (5 %)	1 (8 %)	
Private: fully funded by patient/ family	0	3 (21 %)	2 (9 %)	0	
Health costs charged to patients					
None	28 (76 %)	10 (72 %)	7 (32 %)	4 (31 %)	**<0.001****
Elective care charged to patients	1 (2 %)	1 (7 %)	14 (64 %)	0	
All care charged to patients	8 (22 %)	3 (21 %)	1 (4 %)	9 (69 %)	

3.2. Cranioplasty timeline

The most common timeframe from DC to cranioplasty (35 centres, 41 %) was between 6 weeks and 3 months; followed by between 3- and 6-months after DC (32 centres, 37 %). Nine centres indicated a timeframe of up to 6 weeks post–DC and a further 9 described cranioplasty usually occurring 6 months or more after DC. HIC and LMIC sites showed comparable results ($p = 0.668$).

3.3. Cranioplasty implants

The commonest cranioplasty material used was autologous bone (53 centres, 62 %) and 11 hospitals (13 %), 10 of which were located in LMICs, only implanted autologous bone. Titanium (47 centres, 55 %) and polymethylmethacrylate (45 centres, 52 %) were the commonest artificial implants. Other implants included polyether-ether-ketone (29 centres, 34 %) and hydroxyapatite (21 centres, 24 %).

Regarding artificial implants, 52 hospitals (61 %) used computer planning techniques including computer aided design/computer aided manufacturer (CAD/CAM) to create custom implants for at least some cranioplasties. Of these, 40 (77 %) always outsourced this design process whilst 13 centres (25 %) used in-house CAD/CAM design for some implants, 8 (15 %) doing so exclusively, with HICs and LMICs showing a similar distribution ($p = 0.384$). Twenty-seven centres never used computer planning techniques, with a significant majority being located in 24 being in LMICs ($p < 0.001$). Forty-five (87 %) institutions of those

using implants utilised 3D-printed implants, with most institutions (35) outsourcing 3D-printed implant production and 9 centres (5/9 in HICs) printing implants in their own facility. Two centres located in LMICs used 3D-printed cranioplasty plates supplied via donation.

3.4. Cranioplasty techniques

Intraoperatively, surgical drain insertion was standard practice in 63 centres (73 %) under passive (23 centres, 27 %) or active drainage (40 centres, 47 %), with similar practice in HIC and LMIC groups ($p = 0.382$). Post-cranioplasty antibiotic administration was routine practice in 69 centres (80 %) overall; LMIC sites had significantly higher rates of prophylactic antibiotic use compared to HICs (91 % vs 70 %, $p = 0.02$).

3.5. Bone flap storage

There was variation regarding the fate of bone flap removed during DC, with a near-three-way split between freezing the flap in hospital or private facilities (28 centres); storing the flap in the patient's abdomen (28 centres), and discarding the flap (26 centres) when considering all centres together. When comparing HIC and LMIC sites, LMIC sites were significantly more likely to store the bone flap in the patient's abdomen than HIC sites (25 vs 3 sites), and less likely to discard it (22 vs 44) or store it frozen ($p < 0.001$).

3.6. Barriers to cranioplasty

Overall, the commonest barriers to offering cranioplasty were 1) lack of patient follow-up; 2) cost of plate materials; 3) patient difficulty accessing hospital facilities; 4) operation and hospital stay costs; 5) lack of available plate materials; 6) and patient objection due to religious or ethical principles. LMIC sites reported that operation costs, cost of the implant, lack of cranioplasty plates, and difficulty accessing hospital facilities were more frequent barriers than HIC sites ($p < 0.05$ for all comparisons). Both groups reported lack of follow up and patient objections equally ($p > 0.05$). Additional free text suggestions included: an "overburdened system" with limited availability of operating theatres, patient beds and staff; other surgeries taking precedence resulting in large waiting lists; and patients' relatives opposing cranioplasty, favouring non-surgical management.

3.7. Improving access

Forty-three respondents (50 %) suggested interventions to improve access to cranioplasty. Thirteen suggested reducing costs of materials or care. Ten (9 from LMICs), suggested increasing availability of operating theatres, patient beds, and staff. Additional suggestions included improving access to cranioplasty equipment, 3D printing and clinic services (8 responses, 19 %) and improving availability of materials (7 responses, 16 %). Five responses (12 %) suggested better educating the public about cranioplasty.

3.8. Costs/health economics

Respondents were asked to estimate the costs attributed to cranioplasty. The median (IQR [range]) cost per cranioplasty implant, estimated by 67 institutions (78 %), was 1000 USD (375–4850 [7–20000]), with patients required to cover part of the cost in 14 centres (16 %) and the full cost in 26 centres (30 %). Median cost per procedure, estimated by 72 institutions, was 2000 USD (525–6000 [15–20000]) with patients covering it in full in 25 centres (29 %) and part of it in 15 centres (17 %).

Procedural costs were compared to each country's GNI per capita using World Bank data. Estimated procedure-associated costs were greater than the country's GNI per capita in 7 institutions (8 %) across 6 different LMICs; and more than 50 % of GNI per capita in 13 institutions (15 %) across 11 countries - all but one located in an LMIC. Focussing on cranioplasty materials (excluding procedure-related expenditure), cost was greater than GNI per capita in 3 countries, all LMICs. Centres in Sudan ($n = 1$) and Kenya ($n = 2$) reported the estimated cost of cranioplasty operations (both greater than the country's respective GNI) but did not report material costs alone. Fig. 2 summarises the relationship between GNI and cranioplasty operation (a) and material (b) costs across participating centres.

Estimated procedure costs were also greater than GDP per capita in the same institutions; and more than 50 % of GDP per capita in 14 institutions (16 %), all but one in LMICs.

3.9. Follow-up and ongoing patient care

Eighty-one institutions (95 %) followed-up patients post-cranioplasty, with similar rates in LMICs and HICs (100 % vs 92 %, $p = 0.075$). LMIC sites were more likely to follow patients up within 4 weeks of surgery (71 % vs 36 % in HICs, $p = 0.005$). Follow up in HIC centres was equally split into thirds between the three time windows: within 4 weeks, 4–8 and 8–12 weeks after surgery. In-person appointments were the commonest follow-up modality in all groups (80 sites, 93 %).

For in-patient settings, physiotherapy was the most commonly available service (66 centres, 77 %) followed by speech and language (50 centres, 58 %) and occupational therapy (41 centres, 47 %). Distributions were comparable between HIC and LMIC sites ($p = 0.821$). In community settings following patient discharge, availability of all rehabilitation services declined except for occupational therapy (increased to 44 centres, 51 %) and private rehabilitation services (increased from 9 [11 %] to 17 [20 %] centres).

In 37 centres, DC patients not undergoing cranioplasty lacked access to protective equipment such as helmets. Thirty of these sites were located in LMICs, and LMIC sites had significantly less access to protective equipment compared to HIC sites (30 [77 %] vs 12 [32 %], $p < 0.001$). Twenty-one centres overall stated cranioplasty patients had access to charities or support groups for brain injury, with significantly more access in HICs (15 (54 %) compared to LMICs (6 [13 %]), $p < 0.001$.

Forty-five respondents (52 %) believed stigma affects their patients following DC. Thirty-nine free-text responses described stigma faced by patients. Twenty specified alterations in physical appearance as a source of stigma, whilst 6 attributed it to disability resulting from the medical episode. Two consultant neurosurgeons from separate institutions in Ethiopia described how local beliefs may negatively impact patients:

> "*The community may interpret the defect as something which came following evil doing*", and

> "*People in the countryside may think it's not as a result of trauma rather a spiritual curse that there is a craniectomy defect*".

One neurosurgeon in South Africa highlighted that in younger individuals, head trauma is often stigmatised for its association with "criminal activity":

> *"In younger people, head trauma is often associated with criminal activity, gangsterism"*.

Societal stigma from the inability to return to everyday life and employment was highlighted in multiple responses:

> *"Obvious defect, cosmetically and aesthetically unpleasant until cranioplasty done. Patient cannot freely participate in normal activity"*,

> *"People avoid them because of the missing bone and generally become unemployable as people feel they are a liability"*,

> *"Many times, the patient who has TBI who is recovered and able to work, is rejected for work due to employer fears"*.

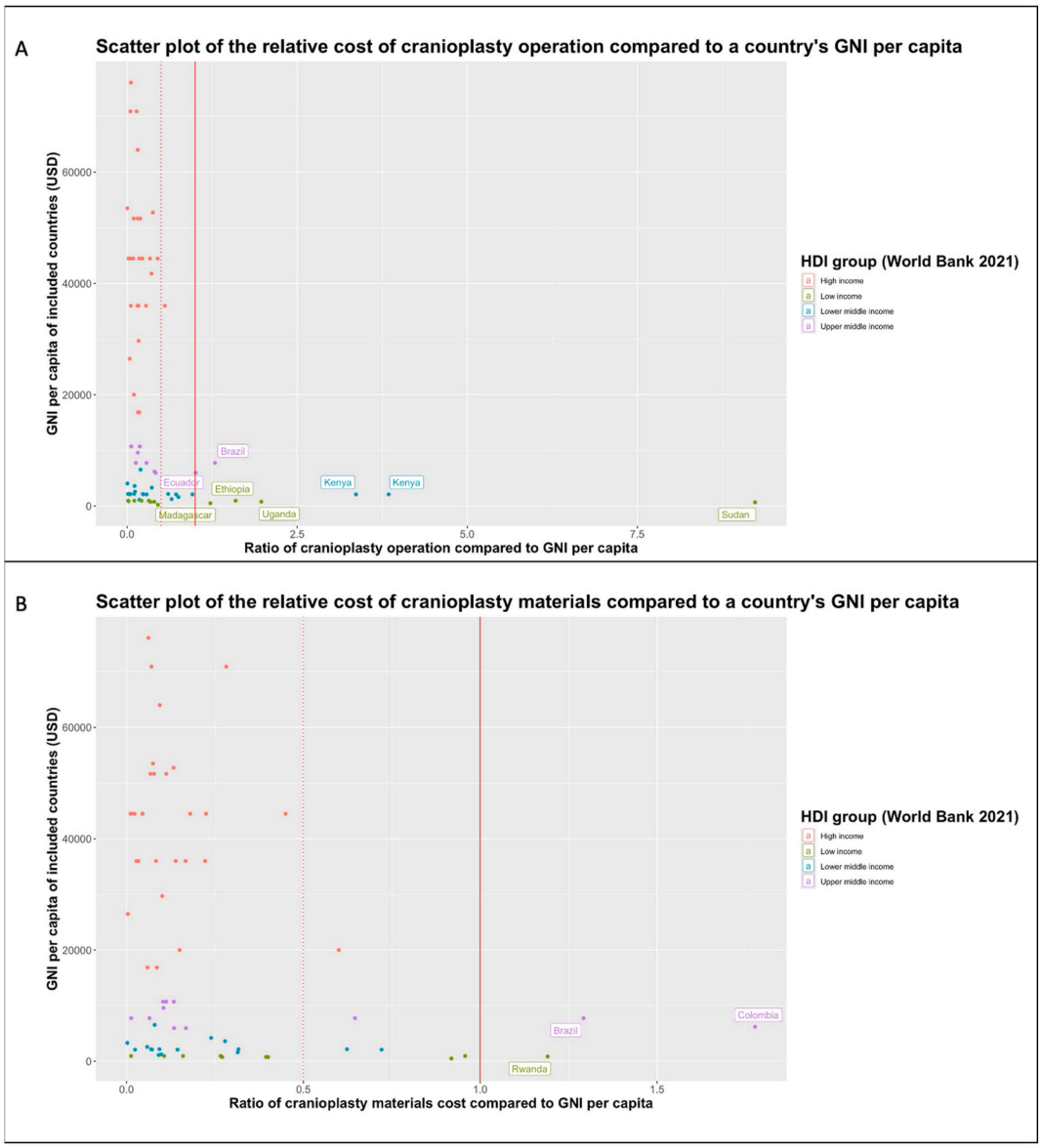

Fig. 2. Scatter plots showing the ratio of cranioplasty (CP) associated costs reported by each centre relative to the country's GNI per capita, grouped by income groups (HIC, UMIC, LMIC, LIC). Each centre is represented by a dot on the chart, income groups are represented by the four colours. Plot A shows the cranioplasty operation/procedural costs, plot B shows the cranioplasty material costs. The vertical lines indicate when the ratio of cranioplasty-related costs to GNI reaches 50 % (ratio = 0.5, dotted line) and 100 % (ratio = 1, solid line). Centres where the ratio of CP/GNI is greater than 1, meaning the cost of CP is greater than the country's GNI, are labelled in both graphs.

4. Discussion

This study aimed to evaluate current global practices around cranioplasty following DC for TBI and identify barriers to performing cranioplasty. The 50-question survey received responses from 86 healthcare centres across 39 countries. Our results, representing one fifth of countries worldwide, are the widest representation of global cranioplasty practice to date.

One key message from the Lancet Commission on TBI is that people in LMICs are disproportionately affected by TBI, with most injuries caused by road traffic incidents. Substantial disparities in care exist, with little infrastructure available for emergency pre-hospital care and little access to post-acute care.[19] Twenty-six countries included in our study (67 %) are classified as LMICs.[20] Concerning participant demographics, information from 78 centres (87 %) was provided by fully qualified neurosurgeons or neurosurgery residents. All other centres were represented by qualified medical doctors. Therefore, it is likely each survey response is a good representation of its institution's cranioplasty practice.

It is not possible to deduce from the data how representative each surveyed centre is of national practice. This would require more data per nation and detailed understanding of the degree of practice standardisation across each nation. Most institutions surveyed were tertiary centres located in urban settings, in line with the specialised nature of cranioplasty practice. Cranioplasty can be classified as a tertiary prevention intervention, facilitating recovery, and is expected to be almost exclusively performed in specialised centres, unlike DC.

Most centres offered cranioplasty to all eligible patients following DC. Lack of infection and good neurological recovery were the commonest indications appearing consistently across LMICs and HICs. Size of the DC flap, on the other hand, was an indication selected predominantly by LMIC centres (31 centres total, 23 located in LMICs) possibly owing to availability of resources limiting maximum cranioplasty size. Most institutions estimated monthly cranioplasty procedure burden at

between 1 and 5 and estimated over 75 % of patients undergoing DC proceeded to cranioplasty. Interestingly, across LMIC sites there were four high-volume centres performing over 10 cranioplasties a month on average. This could be explained by most participating sites being urban referral centres, and neurosurgery care being centralised in many LMICs in few high-volume institutions. The lower numbers of monthly cranioplasties compared to DC could be accounted for by patient mortality from their initial injuries.

Lack of patient follow-up was the highest rated barrier to offering cranioplasty, followed by cost of plate materials; difficulty accessing hospital facilities; operation and hospital costs; lack of available plate materials; and patient objection due to religious or ethical principles. Common barriers equally reported by HIC and LMIC sites were identified (lack of follow up and patient objection to surgery), but costs, availability of implants and access to hospital were more common in LMICs, possibly owing to less resilient health systems. Of the 20 additional barriers suggested by respondents via free text, representatives of 9 centres (across 7 LMICs, 2 HICs) referred to an "overburdened system" in which availability of operating theatres, patient beds and staff limited access, with other surgeries taking precedence. Representatives of 4 centres reported patients' relatives favouring non-surgical management instead of cranioplasty. This highlights areas that could be targeted with future interventions.

Forty-three respondents suggested interventions to improve access to cranioplasty. Key suggestions focussed on reducing costs of materials used or care involved. Nine LMIC respondents suggested increasing availability of operating theatres, beds, and staff. This is in keeping with previous evidence on barriers to surgical care. It is aligned to the findings of the Lancet Commission on Global Surgery, identifying a large gap and inequity in distribution of the surgical workforce in LMICs, setting a target of 100 % of countries with at least 20 surgical, anaesthetic, and obstetric physicians per 100 000 population by 2030.[21]

Improving patient and public education about cranioplasty was also suggested, indicating that both patient and carers' education are important avenues for global policy change to target improved access to cranioplasty.

The most common timeframes from DC to cranioplasty were between 6 weeks and 3 months and between 3- and 6-months. This reflects how recommendations for cranioplasty timing are variable, owing to inconsistent definitions of "early cranioplasty" and mixed results from studies investigating the relationship between timing, complications[10,14,15] and neurological outcome. A recent meta-analysis found no significant difference in complications in relation to timing except that cranioplasty within 90 days of DC was associated with increased odds of hydrocephalus.[9] Despite this, the most common practice for our centres surveyed was cranioplasty within 90 days.

There was considerable variation in first-line practice regarding bone removed during DC, with a near-three-way overall split between extracorporeal cryopreservation in hospital or private facilities, abdominal pocket storage, and disposal. This reflects the available literature, showing that abdominal pocket storage and extracorporeal cryopreservation display similar outcomes.[12] Nearly all hospitals (25/28) where abdominal pocket storage was first line were in LMICs, likely owing to its lower cost (despite requiring additional surgery) relative to cryopreservation or use of artificial implants. This contrasts with HIC hospitals where there was an equal split between cryopreservation (48 %) or discarding the bone flap (48 %) but few sites stored the bone flap in the patient's abdomen (3).

The material used by most institutions for cranioplasty was autologous bone. Twelve hospitals (11 in LMICs) only implanted autologous bone, likely owing to its biocompatibility and low cost. Despite their high cost,[22] titanium and polymethylmethacrylate were the most used artificial implants. The distribution reflects that the optimal material for reconstruction has yet to be agreed upon.[10] A meta-analysis by Henry et al demonstrated the highest revision rate with titanium, whilst polyether-ether-ketone was associated with the lowest risk of re-operation.[11] A subsequent meta-analysis showed risk of infection was 1.62 times higher when polymethylmethacrylate was used compared with autologous bone and that length of stay after titanium cranioplasty was on average 3.62 days shorter compared with autologous bone.[23] Authors concluded "there is not strong enough evidence available to make comprehensive conclusions regarding the risk profiles of various cranioplasty materials across multiple outcomes".[23]

Regarding CAD/CAM design of implants, significant differences in access were seen between HICs, where support was often used, and LMICs where this was available less than 50 % of the time. It is highly likely the number of centres in LMICs utilising 3D-printing technology will drastically increase in the future due to the emergence of personal desktop 3D-printers and open-source software lowering costs.[22,24,25] However, the most cost-effective cranioplasty implant remains to be determined with mixed results from studies to date.[26,27]

When investigating costs and economic impact associated with cranioplasty procedures, we observed that needing a cranioplasty operation can result in impoverishing or catastrophic expenditure for patients. Respondents were asked to estimate the costs attributed to cranioplasty in their centres. Of the 67 institutions providing data on the cost of cranioplasty implants, 14 centres stated that patients cover at least part of the cost, and 26 that patients cover the full cost. Similar results were seen for the cost of a cranioplasty operation. These estimated costs were greater than the country's GNI per capita in 7 institutions (all LMICs); and greater than 50 % of GNI per capita in 13 institutions, all but one in a LMIC. Despite the small sample size, this data from several countries suggests the cost of cranioplasty can be incredibly high making it unaffordable, especially where patients and families are responsible for funding elective surgical care. In a few settings, cranioplasty costs were more than double the country's GNI. The variation seen may be influenced by income levels within the region and differences in financing methods for healthcare. Patients had to pay out-of-pocket to cover entire procedural costs in 4 of 7 centres and part of the procedural costs in 2 of 7 centres. This is particularly important considering the Lancet Commission on Global Surgery highlighted protection against impoverishing expenditure (out-of-pocket payments pushing one into poverty or further into poverty) and catastrophic expenditure (out-of-pocket payments greater than 40 % of household income net of subsistence needs) as core indicators for monitoring universal access to affordable surgical care, setting targets of 100 % protection against both by 2023.[28] Whilst assessing whether such targets have been met is beyond the scope of this article, it shows there is a high likelihood of impoverishing or catastrophic expenditure for patients, especially as non-medical costs of accessing care (e.g. transportation) are not included here. However, it is important to highlight that costs were estimated by respondents and that future research with objective measurement of costs is warranted to support our findings.

It is important to remember cranioplasty is not the end of a TBI patient's care journey. Most institutions (95 %) followed-up patients post-cranioplasty, the majority doing so within 3 months of discharge. 93 % followed-up patients via in-person appointment, a portion which could have since reduced due increased virtual physician–patient interactions during the COVID-19 pandemic. Longer term follow-up beyond 3 months is relevant as cranioplasty complications (e.g. implant displacement, fracture) can occur after a longer interval. It would be important to develop strategies to improve longer-term follow-up in resource limited settings, for example utilising smartphone technologies, overcoming barriers such as distance from hospital. TBI affects multiple domains of functioning, and outcomes are affected by personal characteristics and life-course events, consistent with a multifactorial bio-psycho-socio-ecological model of TBI.[19] Access to rehabilitation services following TBI is inconsistent and no evidence-based protocols for treating long-term problems (including fatigue, cognitive and behavioural changes) currently exist.[19] There are reports of unmet rehabilitation needs and long-term service use below the prevalence of morbidity - with insufficient care in both HICs and LMICs.[19,29–31]

Furthermore, research suggests rehabilitation referral following TBI is not only influenced by clinical factors but also demographic, social, and organisational factors.[19] Concerning rehabilitation services offered post-cranioplasty, our data show physiotherapy was offered the most, followed by speech and language therapy and occupational therapy. This is a similar trend to that seen for individuals enrolled in the CENTER-TBI study.[29]

Less than a quarter of surveyed sites overall stated cranioplasty patients had access to charities or support groups for brain injury. This number was 54 % in HIC and 13 % in LMIC sites. This area could be targeted to improve access to support networks for patients. In addition, over 50 percent of respondents said that they think stigma affects their patients following DC, with commonly reported sources of stigma being alterations in physical appearance and disability resulting from the medical episode. Respondents highlighted the role of local culture and beliefs, suggesting rural communities may perceive patients with craniectomy defects as victims of a "spiritual curse" or as something acquired through "evil doing". Social contexts also play a role, for example by associating head trauma with "criminal activity" in some settings. Awareness of cultural and societal values in different contexts is essential to target advocacy and education campaigns to improve awareness at community level.

5. Limitations and future recommendations

While centres from several countries have been sampled; the dataset is not wholly representative of global practice. The survey also introduces bias as responses were obtained from a self-selected cohort where respondent volunteered to complete the survey. Furthermore, respondents provided estimates based on their local knowledge. Such estimates may have been skewed due to the impact of the COVID-19 pandemic on services during the period when the survey was open. Also, whilst efforts were made to limit the number of survey questions to 50, it is possible that decision fatigue could have impacted responses and estimates, particularly in later questions.

Further cross-sectional research accurately quantifying cranioplasty practice is warranted. Current cranioplasty databases are lacking, even in HICs, with the United Kingdom only recently establishing one and reporting findings.[32,33] To effectively impact clinical care and outcomes, interactions are also needed with research funders, regulators, policy makers, and patient organisations.[19] Given the barriers identified, alternative surgical approaches not requiring cranioplasty should be investigated. For example, hinge craniotomy, where cranial decompression is achieved by loosely replacing the bone flap at the end of the surgical procedure, has been utilised in a variety of settings.[34] Systematic evaluation is needed to identify scenarios where hinge craniotomy offers the same clinical benefits, with fewer side effects compared to DC and cranioplasty. In resource-limited settings this may be particularly beneficial in the longer-term by avoiding complications of craniectomy and cranioplasty which can result in additional unaffordable costs.

When DC is performed, exploring the role of alternatives to cranioplasty including protective helmets should also be investigated formally, as these may be more accessible and affordable than further operations. Our responses showed that access to protective equipment is still limited, inviting work to assess innovative ways of improving this. Similarly, qualitative research should focus on how different interventions can be applied across cultural and societal contexts, considering how factors such as stigma and beliefs around outcome can influence the role of interventions such as cranioplasty.

6. Conclusion

This survey-based study aimed to increase the currently limited evidence around cranioplasty following TBI. Through surveying practitioners across a HICs and LMICs, this survey described the current state of cranioplasty practice globally, highlighting variations in clinical practice and identifying key barriers preventing patients from accessing this service. This can drive future work to target interventions addressing these barriers, building surgical capacity and equitable access to surgery following TBI.

CRediT authorship contribution statement

Sara Venturini: Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Saniya Mediratta:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Tobias J. Adams:** Writing – original draft, Visualization, Formal analysis, Data curation. **Harry Mee:** Writing – review & editing, Conceptualization. **Bhagavatula Indira Devi:** Writing – review & editing. **Tariq Khan:** Writing – review & editing. **Andres M. Rubiano:** Writing – review & editing. **Franco Servadei:** Writing – review & editing. **Adel Helmy:** Writing – review & editing. **Angelos Kolias:** Writing – review & editing. **Peter J. Hutchinson:** Writing – review & editing, Supervision.

Declaration of competing interest

Declarations of interest: none.

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Appendix A. Supplementary data

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Abbreviations

CAD/CAM: (computer aided design/computer aided manufacturer)
CP: (cranioplasty)
DC: (decompressive craniectomy)
GDP: (gross domestic product)
GNI: (gross national income)
HICs: (high-income countries)
IQR: (interquartile range)
LMICs: (low- and middle-income countries)
NIHR: (National Institute for Health and Care Research)
TBI: (traumatic brain injury)
USD: (U.S. Dollar)